Proof: Observe that $\gcd(f,g)=0 \iff f=0$ and $g=0$. We can assume that $f$ or $g$ is not $0$, for otherwise the assertions are clear. Fix $h \in \langle f,g\rangle \setminus \{0\}$ of smallest degree. As discussed in the previous proof, $\langle h\rangle = \langle f,g\rangle$. We claim that $h=\gcd(f,g)$. So, let $p \in k[x]$ be any polynomial that divides ~~both~~ $f$ and $g$. That means, $f = Cp$ and $g = Dp$ for some $C, D \in k[x]$.
On the other hand, since $h \in \langle f,g\rangle$, one has $h = Af + B\cdot g$ for some $A, B \in k[x]$. $\Longrightarrow$

$$h = Af + Bg = ACp + BDp = (AC+BD)p \Longrightarrow p \text{ divides } h.$$

So, every divisor of $f$ and $g$ divides $h$. And $h$ is a divisor of $f$ and $g$, because from $\langle f,g\rangle \subseteq \langle h\rangle$, we know that $f \in \langle h\rangle$ and $g \in \langle h\rangle$, so both $f$ and $g$ are divisible by $h$. Thus, $h = \gcd(f,g)$.

Let's verify the <u>uniqueness</u> of $\gcd(f,g)$ up to a factor in $k\setminus\{0\}$. Let $H$ be any greatest common divisor of $f$ and $g$. Because $h$ is a divisor of $f$ and $g$, we have $h = q\cdot H$ for some $q \in k[x]\setminus\{0\}$.
So, $\deg h \geq \deg H$. On the other hand, symmetrically, since $h$ is a greatest common divisor and $H$ is a divisor of $f$ and $g$, we also have $H = Q\cdot h$ for some $Q \in k[x]\setminus\{0\}$. So,
$\deg h \leq \deg H$. $\Longrightarrow \deg h = \deg H \Longrightarrow h = qH$ with $\deg q = \deg H - \deg h = 0$.

Algorithm to find gcd (f). If $g=0$, then $\gcd(f,g)=\gcd(f,0)=f$.

Assume $\deg f \geq \deg g$. Then one can do long division of $f$ by $g$ obtaining $f = q\cdot g + r$ with $q, r \in k[x]$ and $\deg r < \deg g$. It turns out that

$$\gcd(f,g) = \gcd(r,g),$$

for $r = f - q\cdot g$. If a polynomial $p \in k[x]$ divides $f$ and $g$, then $p$ also divides $r$ and $g$. The other way around, if a polynomial $p \in k[x]$ divides $r$ and $g$, then $p$ also divides $f$ and $g$, because $f = q\cdot g + r$.

The gain is $\underbrace{\gcd(f,g)}_{\text{the minimum of the degrees of } f \text{ and } g \text{ is } \deg g.}$ vs. $\underbrace{\gcd(r,g)}_{\text{the one with the smaller degree is } r \text{ and its degree is } < \deg g.}$

As a pseudocode:

```
Greatest_Common_Divisor (f, g):
    while g ≠ 0:
        ▷ Invariant: we don't change gcd(f,g), while changing (f,g)
        q, r := Univariate_Division(f, g)   ▷ f = q·g + r with deg r < deg g.
        ▷ We know: gcd(f,g) = gcd(g,r)
        f, g := g, r   ▷ the degree g gets smaller - progress towards having g=0.
    return f
```

The algorithm is calle the Euclidean algorithm. Remark: one also has an extended version of this, called the extended Euclidean algorithm. the extended version provides $A, B \in k[x]$ with $\gcd(f,g) = Af + Bg$, that means it provides the evidence in the form of $A$ and $B$ that $\gcd(f,g)$ belongs to the ideal $\langle f, g \rangle$.

$\square$

**Def 1.5.8.** A greatest common divisor of $f_1, \dots, f_s \in k[x]$ is a polynomial $h \in k[x]$ with the following properties.

(i) $h$ divides $f_1, \dots, f_s$ (all $s$ polynomials)

(ii) every polynomial dividing all of the $f_1, \dots, f_s$ also divides $h$.

Notation: $h = \gcd(f_1, \dots, f_s)$.

**Prop 1.5.9.** Let $f_1, \dots, f_s \in k[x]$ with $s \geq 2$. Then the following are true.

(i) $\gcd(f_1, \dots, f_s)$ is unique up to multiples in $k \setminus \{0\}$.

(ii) $\langle \gcd(f_1, \dots, f_s) \rangle = \langle f_1, \dots, f_s \rangle$

(iii) for $s \geq 3$, then $\gcd(f_1, \dots, f_s) = \gcd(f_1, \gcd(f_2, \dots, f_s))$.

(iv) There is an algorithm that determines $\gcd(f_1, \dots, f_s)$ for given $f_1, \dots, f_s$.

*Proof*: exercise.

**Corollary 1.5.10.** There is an algorithm deciding if $f$ belongs to $\langle f_1, \dots, f_s \rangle$ for given $f_1, \dots, f_s, f \in k[x]$.

GreatestCommonDiv$(f_1, \dots, f_s)$:

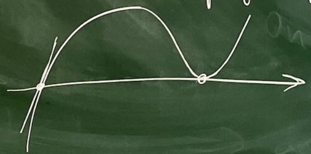
TWO IDEAS

*Proof.* Determine $g = \gcd(f_1, \dots, f_s)$.

To decide if $f \in \langle f_1, \dots, f_s \rangle = \langle g \rangle$

By long division of $f$ by $g$ decide if $f \in \langle g \rangle$ is true. $\square$

## Exercises 1.5.11

1. Do long division of $x^3+2x^2+x+1$ by $2x+1$ $(k=\mathbb{R})$.
2. $\gcd(x^4-1, x^6-1) = ?$
3. Find an algorithm that determines $A, B \in k[x]$ with $\gcd(f,g) = Af + Bg$, where $f, g \in k[x]$. Understand why it's a correct algorithm.
4. Assume that we want to see if $V_k(f_1, \dots, f_s) \neq \emptyset$ in the univariate case $f_1, \dots, f_s \in k[x]$. How do we solve this task? If $k$ is not algebraically closed, this can be tricky. So, assume $k$ is algebraically closed, i.e., $k=\mathbb{C}$ (if you prefer cryptography take $k$ to be the algebraic closure of a finite field!) Can you use what's been discussed (gcd), to solve or at least simplify this problem.
5. 

   A root $r$ of $f \in k[x]$ is said to be a multiple root if $f(r)=0$ and $f'(r)=0$. How to decide if $f \in k[x]$ has a multiple root, say, $k=\mathbb{C}$?

⑥ Transfer the contents of this section from the ring $k[x]$ to the ring $\mathbb{Z}$.

| | |
|---|---|
| $k[x]$ | $\mathbb{Z}$ |
| $f \in k[x]$ | $z \in \mathbb{Z}$ |
| $\deg f$ | $\lvert z \rvert$ |
| $k \setminus \{0\}$ | $\{1, -1\}$ |

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## $\boxed{1.6}$ Residue class rings

ring := unitary commutative ring.

**Def 1.6.1** A map $\varphi : R \to S$ from a ring $R$ to a ring $S$ is called a ring homomorphism if

if $\varphi(a+b) = \varphi(a) + \varphi(b)$, $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$ for all $a, b \in R$ and $\varphi(0) = 0$, $\varphi(1) = 1$